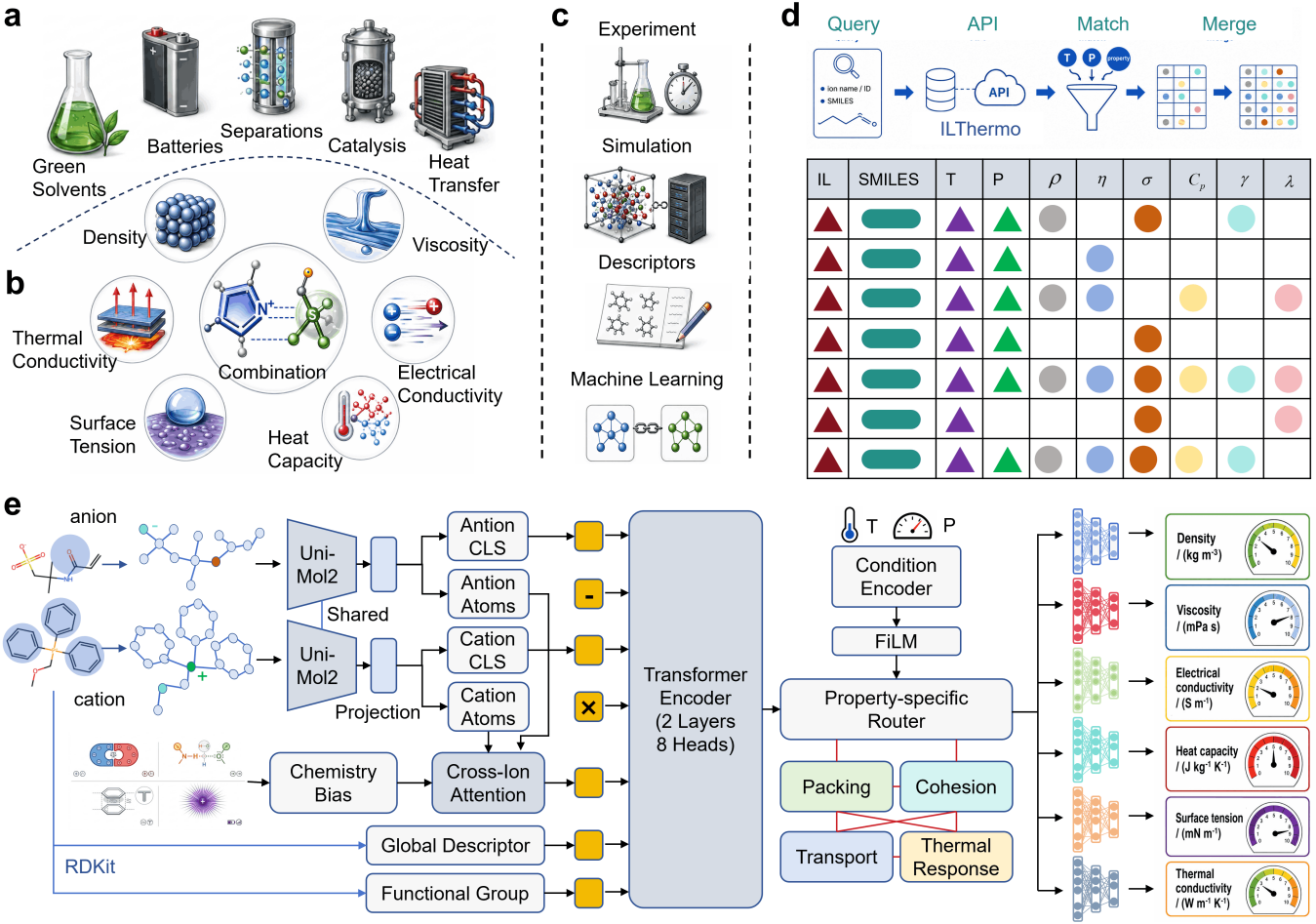

a
Green Solvents
Batteries
Separations
Catalysis
Heat Transfer
Density
Viscosity
b
Thermal Conductivity
Combination
Electrical Conductivity
Surface Tension
Heat Capacity
c
Experiment
Simulation
Descriptors
Machine Learning
d
Query
API
Match
Merge
ILThermo
IL
SMILES
T
P
e
anion
cation
Uni-Mol2
Shared
Uni-Mol2
Projection
Anion CLS
Anion Atoms
Cation CLS
Cation Atoms
Chemistry Bias
Cross-Ion Attention
RDKit
Global Descriptor
Functional Group
Transformer Encoder (2 Layers 8 Heads)
Condition Encoder
FiLM
Property-specific Router
Packing
Cohesion
Transport
Thermal Response
Density / (kg m-3)
Viscosity / (mPa s)
Electrical conductivity / (S m-1)
Heat capacity / (J kg-1 K-1)
Surface tension / (mN m-1)
Thermal conductivity / (W m-1 K-1)